Bracket Sequences I

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2064 | Mathematics | https://cses.fi/problemset/task/2064

Your task is to calculate the number of valid bracket sequences of length n. For example, when n=6, there are 5 sequences:

- ( ) ( ) ( )
- ( ) (( ))
- (( )) ( )
- ((( )))
- (( ) ( ))

Input

The only input line has an integer n.

Output

Print the number of sequences modulo 10^9+7.

Constraints

- $1 \le n \le 10^6$

Example

Input
6

Output
5